On Field Performance Analysis of IEEE 802.11p and WAVE Protocol Stack for V2V & V2I Communication

Vivek N,
Team Embedded
Centre for Development of Advanced Computing (C-DAC),
Hyderabad, India,
vivekn@cdac.in

S.V.Srikanth,
Team Embedded
Centre for Development of Advanced Computing (C-DAC),
Hyderabad, India,
venkatas@cdac.in

Saurabh P,
Team Embedded
Centre for Development of Advanced Computing (C-DAC),
Hyderabad, India,
aurabhp@cdac.in

T.P.Vamsi,
Team Embedded
Centre for Development of Advanced Computing (C-DAC),
Hyderabad, India,
vamsip@cdac.in

Raju K,
Team Embedded
Centre for Development of Advanced Computing (C-DAC),
Hyderabad, India
rajukundhe@aol.com

***Abstract*— The Automotive Industry worldwide is considering Dedicated Short Range Communication (DSRC) as a promising technology for Vehicle to Vehicle (V2V) and Vehicle to Infrastructure (V2I) communication to support public safety applications, improvement of traffic flow and a number of ITS applications. Though the research on the current topic has been extensively done, yet it is not fully evaluated and analyzed through on field experiments. These wireless networks are prone to fading due to shadowing from obstacles or multipath propagation resulting in attenuation of signal at receiver whichleads to high error rates. In this paper we describe and present the results of field experiments that were conducted to evaluate various wireless communication parameters over 802.11p based WiFi network using IEEE 1609 WAVE protocols for a dedicated DSRC hardware. The parameters include delay,jitter and packet loss in DSRC based system when it is made to communicate under various channel load conditions in outdoor environment. We also evaluate and present the results of communication performance under cross channel interference in real field environment.**

Keywords— DSRC, WAVE, Vehicular Communication, V2V, V2I, IEEE 802.11p

I. INTRODUCTION

Dedicated Short Range Communication (DSRC) is an emerging medium range wireless technology aimed primarily at vehicular safety. The Automotive Industry has started working to develop DSRC technology for use in Vehicle to Vehicle (V2V) and Vehicle to roadside Infrastructure (V2I) communication. For this purpose, US Federal Communications Commission (FCC) has allocated 75 MHz of licensed spectrum at 5.9GHz for the operations of Intelligent Transportation System (ITS) [8]. Other countries including Japan, Europe, and Korea also recognized the importance of DSRC but the spectrum is structured differently. For India, the spectrum under consideration for DSRC is from 5.725GHz-5.825GHz with ten channels of 10MHz bandwidth[18].

Various researches in the past have been carried out to evaluate 802.11p performance in V2V and V2I based applications. But most of the previous research has used simulators [1][2][3] to model the communication over IEEE802.11p. However the real worldenvironment is dynamic,it brings far more challenges. Therefore it becomes necessary to evaluate and verify these communication metrics practically.

A. Paier et al [4] carried out infrastructure to vehicle trial to evaluate wireless performance. They were able to record how environmental effects and shadowing by moving vehicle leads to high fluctuations. It was also seen that antenna patternwas strongly influenced by neighboring structure. In their tests they were able to infer that high communication range achieved lower data rate.

G P Grau et al [6] made empirical analysis in evaluation of IEEE802.11p radio channel in a different vehicular environment. K Karlsson et al [5] and A Bohm et al [7] explored the effectsof V2V communication in different traffic scenarios. They observed vehicle speed, direction of motion, Line of Sight and Non line of Sight situations by a moving vehicle in suburban and highway environment.

Here in this paper we investigate DSRC performance by measuring changes in application to application level delay, jitter and packet loss that is experienced through our experimentation. The communication was doneusing WAVE Short Message Protocol (WSMP) over 802.11p network under various channel load condition in an outdoor environment. These tests help us measure the baseline performance that can be achieved in real conditions and help us design better V2V and V2I applications.

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Remainder of the paper is organized in the following manner, overview of DSRC technology is discussedin section 2. Section 3 reveals our experimental methodology, which we followed to gain insight into the DSRC communication performance in outdoor conditions. Section 4 throws light on the results of the experiments conducted.We conclude with our experiences on DSRC in section 5 and finally highlight the future work intended to do in this field.

II. DSRC OVERVIEW

DSRC standards are based on 802.11a, with adjustments made for low overhead operations in 5.9GHz. 802.11p Wireless Access for Vehicular Environment (WAVE) is the amendment that allows wireless devices to communicate with each other in a high speed mobility vehicular environment. For vehicular safety it is an advantage to operate at 5.9GHz band because it avoids the overcrowded band used by other wireless devices. Hence DSRC medium access is subjected to less competition against other wireless stations and gives a high probability for message transmission in vehicle safety critical applications. High data rate communication is offered by this band, but high frequency operation imposes some disadvantages.They have low penetration capability as compared to low frequency signals which can reduce communication range in urban environments [12].

There are two types of channels available in DSRC namely Control Channel (CCH) and Service Channel (SCH) as shown in Fig.1. All the channels have 10MHz of bandwidth. The CCH is for safety critical applications and SCH can be used for both safety and non-safety applications such as infotainment [8].

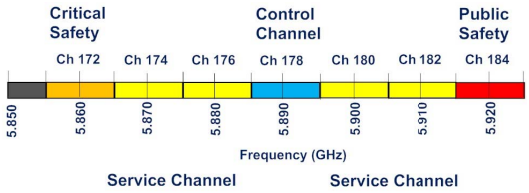


Fig.1 DSRC Spectrum

The WAVE stack for DSRC is shown in the Fig.2. It defines complete standards, services and functions that a station should perform ina vehicular network. The stack is structured into two planesi.e. data plane and management plane. The WAVE subsystem is managed by software element known as WAVE Management Entity. It is responsible for handling configurations, management frames and service advertisement.

DSRC applications can exchange data using WSMP protocol or traditional UDP/TCP protocols that are located in data plane. WAVE Short Message Protocol (WSMP) is specified in 1609.3 standards [10]. WAVE Short Messages are used for broadcast applications that require instant exchange of safety information between vehicles or Road Side Unit (RSU). TCP/UDP runs on top of Internet Protocol Version 6 (IPV6)and are used for Internet connectivity. Data Link layer is divided into LLC(Logic Link Control), upper and lower MAC. LLC is defined by IEEE 802.2 and provides standard interface to lower layer. WAVE upper MAC is defined by IEEE 1609.4 which describes multichannel wireless operation and channel coordination [9]. IEEE 802.11p comprises of lower MAC and PHY layers. Within the WAVE stack IEEE 1609.2 provides security services and privacy protection mechanisms.This defines the secure message formats and processing of secured messages [11].

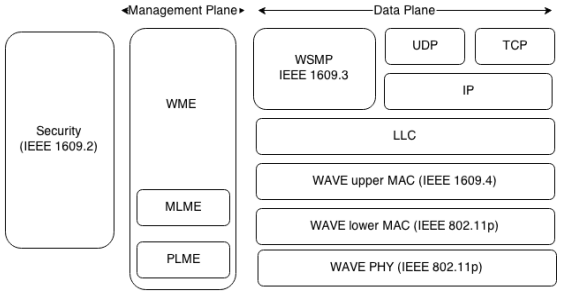


Fig.2 WAVE Protocol Stack

III. EXPERIMENTAL METHODOLOGY

In this section we describe our experimental methodology and the testbedsetup to collect various parameters. This will help the readers to know how systematic tests were conducted and the respective procedures that were adopted.

A. Hardware & Software features

The hardware package of On Board Unit (OBU) and Road Side Unit (RSU) contains MIPS processor running at 680MHz, Flash – 16MB, SDRAM – 64MB, Gigabit Ethernet interface, USB interface, high power Atheros AR5414 based WLAN card and Integrated GPS device, with RF antenna.



Fig.3 OBU & RSU Hardware

Both the unitsi.e. OBU & RSU shown in Fig 3 have WAVE standard support for 802.11p, 1609, SAE J2735. The respective units operate at the 75MHz license band in 5.9 GHz spectrum with 7 channels of 10 MHz channel bandwidth, multichannel operation support (control channel and service channel) using single radio, GPS based synchronization, high data rate up to 27Mbps operation in 10MHz channel width.

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They support Linux 2.6 Kernel and respective API's to develop WAVE applications [15][16].

The difference between OBU and RSU is that RSU hasan additional WiFi radio working at 2.4 GHz ISM band. The WiFi radio can be used to connect to back-haul network. In addition RSU resides in a weather proof IP67 rated enclosure as it is normally deployed outdoor and has to withstand harsh environmental conditions as compared to OBU that sits inside a vehicle

B. Experimental setup

The experimental setup consists of two OBU and one RSU. In the analysis we were focused to record the communication performance between OBU-RSU and OBU-OBU. Parameters of interest include average round trip delay, jitter and percentage of WAVE Short Message Protocol (WSMP) packet lost. The test site chosen for conducting the tests was a 600 meters distance road located atShamshabad in Hyderabad. RSU was mounted on a road side pole at a height of 6meters.One OBU was kept on the top of vehicle. Second OBU was kept near RSU and was used to generate traffic in the communication channel and adjacent channel. This aided us to measure the effect of channel load and cross channel interference. All the three units were connected via Internet so that necessary scripts and test programs could be run remotely. The architecture adopted for the test is shown in Fig 4.

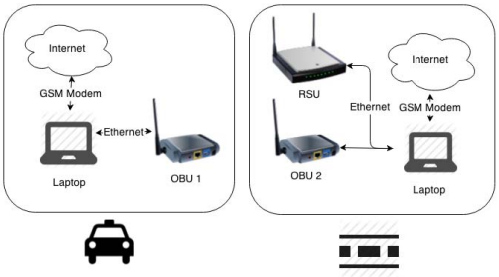


Fig.4Testbed Setup

All the set of experiments were performed for CCH 178 and SCH 172 at varying distance from RSU at 6Mbps data rate. The time on the units were synchronized to GPS clock. For testing, custom scripts were developed.An application wasalso developed for OBU that transmitted data packets using Wave Short Message Protocol(WSMP) at 0.1 Hz with packet size of 400bytes including the time stamp on a single channel.Another application running on RSU which was listening on the same channel received the packet and retransmitted them again to the respective OBU. The data rate chosen for both the channels was 6Mbps. Channel switching was not performed between CCH and SCH.The units were made to transmit on the single radio channel during the tests. Using this procedure parameters like average delay, average inter-arrival jitter and packet loss were calculated in the test setup. The same tests were performed with second OBU generating traffic in the same and adjacent channel along with varying load.

The testing was performed on a 600m stretch of a road isolated from regular traffic and readings were recorded at 100m interval. Google map view of testing sight is shown in the Fig.5. The road was straight, but there was a dip of about 1.5m as depicted in the Fig.6. This accounted for absence of Line of Sight (LOS) between RSU and OBU from 180m to 450m of the road stretch.



Fig.5 Test Site

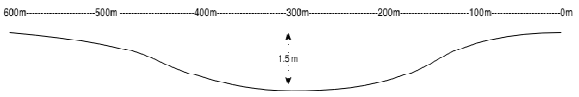


Fig.6 Road Stretch

IV.RESULTS

Though the tests were conducted for communication between both RSU-OBU and OBU-OBU, here we only present result of RSU-OBU as both the tests produced almost similar results.

From the graph shown in the Fig.7, percentage of packet loss increases as the distance between the RSU and OBU was increased for both the channel. But there was sudden rise in packet loss as shown in the graph after 200m. This we believe was because of environmentalfeatures that degrade the transmission quality between the two units predominantly the dip in the road which accounted for absence of LOS.

From Fig 8 and Fig 9 we can however see that there was no considerable change recorded in average delay and jitter with increased distance. Average delay was between 2 to 3 milliseconds and average inter-arrival jitter was within 0.4 to 1.3 milliseconds for both the CCH and SCH channel. Delay [17] and jitter calculation were done according to formulas mentioned in equation 1 and 3.

Average delay = Sum of round trip time of packets received / Total number of packets received (1)

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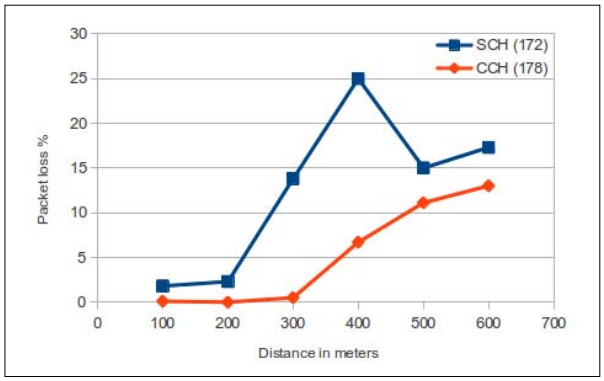


Fig.7 Packet loss

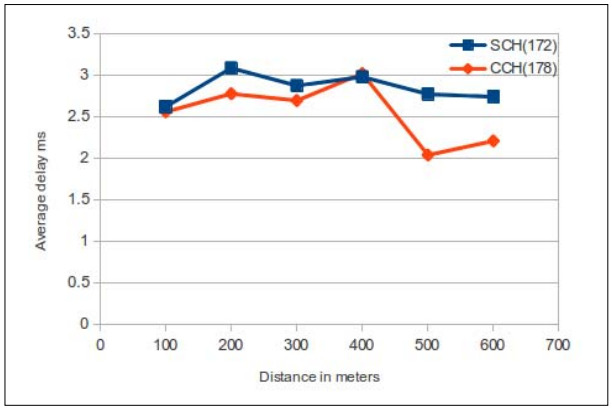


Fig.8 Average delay

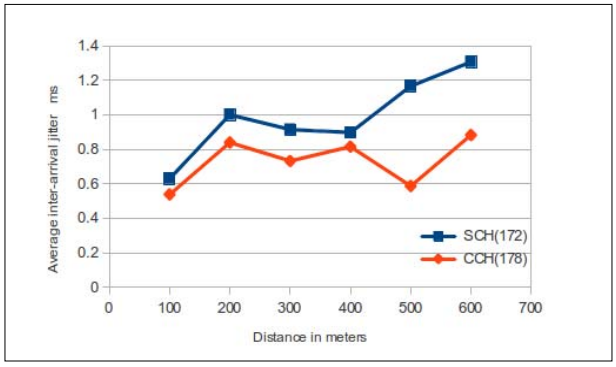


Fig.9 Average Inter-arrival jitter

"Jitter J is defined to be the mean deviation of the difference D in packet spacing at the receiver compared to the sender for a pair of packets. If Si is the time-stamp from packet i, and Ri is the time of arrival in time-stamp units for packet i, then for two packets i and j, D may be expressed as

$$D(i,j)=(Rj-Ri)-(Sj-Si)=(Rj-Sj)-(Ri-Si) \quad (2)$$

The inter-arrival jitter is calculated continuously as each data packet i is received from source SSRC_n, using this difference D for that packet and the previous packet i-1 in order of arrival, according to the formula" [14]

$$J=J+(|D(i-1,i)|-J)/16 \quad (3)$$

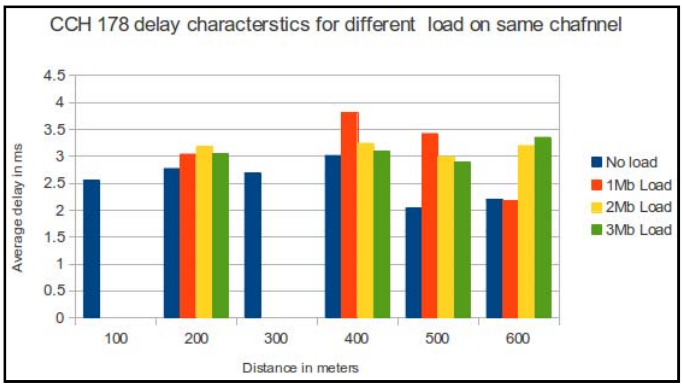
Tests were also conducted to record delay, inter-arrival jitter and packet loss % when communication channel is subjected to varying load. Load of 1Mb, 2Mb and 3Mb was generated using second OBU kept near RSU. The effects of cross channel interference were also measured by generating traffic in adjacent channel. Figure 10 shows the full details of recorded results for CCH(178).

V. CONCLUSION

We evaluated the outdoor performance of off-the-self DSRC enabled platform. To our findings we infer that the average delay experienced by WSMP packets was less and did not change drastically irrespective of varying channel load on the same channel. We could observe variations for jitter. The packet error that accounted for loss of packets varied a lot with distance as a direct result of increase in channel load, cross channel interference. Line of Sight and the environment around OBU seems to have a major impact on communication. In our case varying load on adjacent channel had a lot of effect in terms of cross channel interference and resulted in high packet loss.

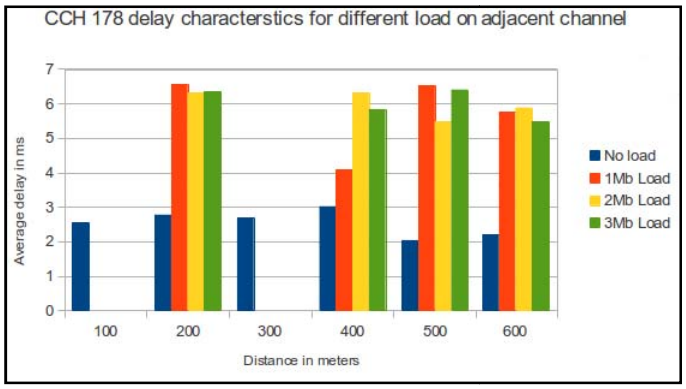
VI. FUTURE WORK

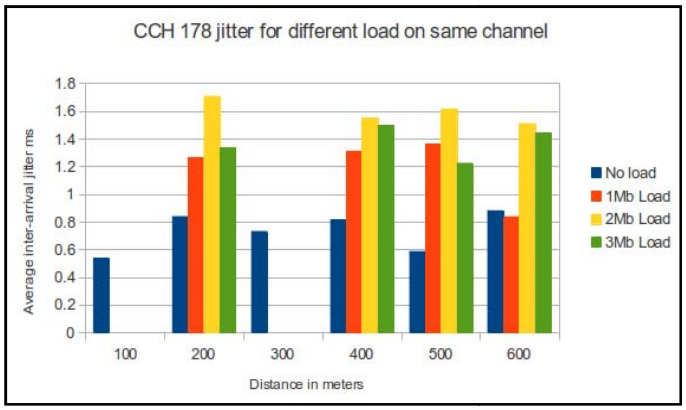
With our current understanding of WAVE protocol stack we intend to develop a testbed for V2V and V2I safety application using DSRC technology. Presently we are looking into V2I based application that issues advisory, roadside alerts and speed limit audio visual warnings to driver. We are also looking forward towards the design &development of customizedcost effective DSRC enabled hardware platform with WAVE stack.

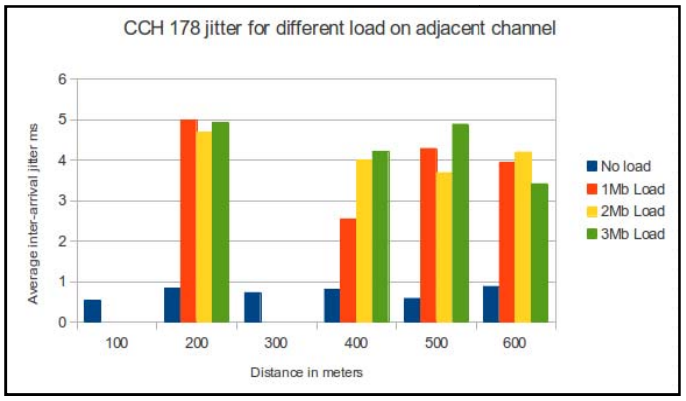


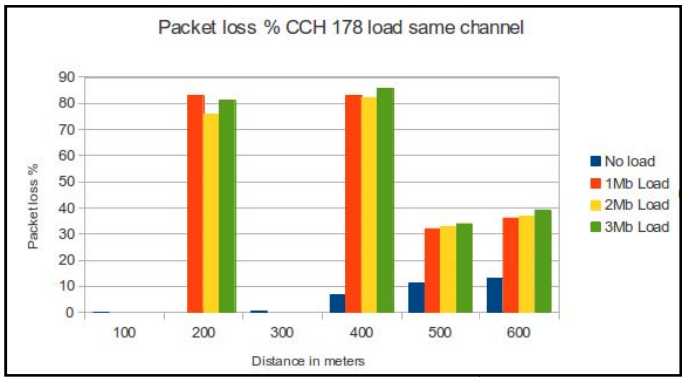
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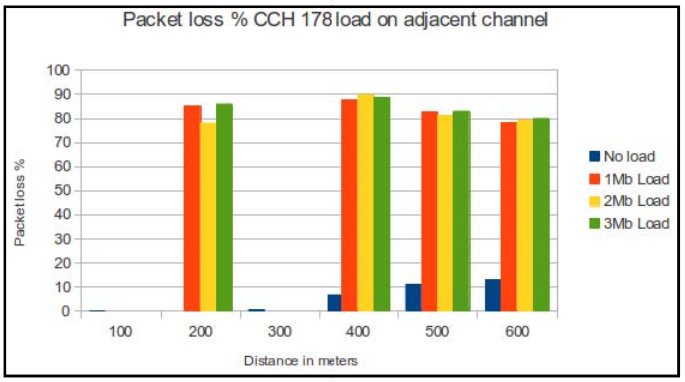


Fig.10 Overall Results for CCH (178)

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